

Database and Collection Management (Create, View, Rename, Delete)

1. Introduction to MongoDB Structure

1.1 What Is a Database?

A **database** is a container that stores organized data on the server. Example scenario:

- A school database may contain student records, library records, and department information.

1.2 Collections

Inside a database, MongoDB stores data in **collections**. A collection is similar to a file or table that holds related records.

Examples in a school database:

- `students`
- `library`
- `teachers`

1.3 Documents

A **document** is a single record inside a collection. MongoDB documents are in **BSON/JSON-like structure**.

Example:

```
{
  "name": "John",
  "age": 20,
  "class": "10A"
}
```

1.4 Hierarchy Summary

```
Database
├── Collections
│   └── Documents
```

2. Connecting to MongoDB Shell

MongoDB operations are executed using the **Mongo Shell (mongosh)**.

Methods to open mongosh

1. **Via installation folder** Double-click `mongosh.exe` inside the MongoDB installation directory.

2. **Via system command prompt**

```
mongosh
```

Once opened, MongoDB displays a successful connection message.

3. Viewing Existing Databases

Command

```
show dbs
```

This lists all available databases such as:

- `admin`
- `config`
- `local`

Note: Newly created databases appear only after at least one collection or document is added.

4. Creating and Selecting a Database

Command

```
use databaseName
```

Example

```
use school
```

MongoDB behavior:

- If the database exists → it switches to it.
- If not → it creates the database and switches to it.

Check current database

```
db
```

5. Creating a Collection

Syntax

```
db.createCollection("collectionName")
```

Example

Create a **students** collection:

```
db.createCollection("students")
```

Response:

```
{ "ok" : 1 }
```

View all collections

```
show collections
```

Example output:

```
students  
library
```

6. Renaming a Collection

Syntax

```
db.oldCollectionName.renameCollection("newCollectionName")
```

Example

Rename `students` → `student`:

```
db.students.renameCollection("student")
```

Verification:

```
show collections
```

7. Helpful MongoDB Shell Commands

7.1 General help

```
help
```

7.2 Database-level help

```
db.help()
```

Displays commonly used operations such as:

- `find`
- `count`
- `updateOne/updateMany`
- `deleteOne/deleteMany`
- `createCollection`
- Indexing methods

7.3 Collection-level help

```
db.collectionName.help()
```

Example:

```
db.student.help()
```

Shows commands specifically usable on that collection.

8. Deleting a Collection

Syntax

```
db.collectionName.drop()
```

Example

```
db.library.drop()
```

If successful, MongoDB returns:

```
true
```

9. Deleting a Database

Before deleting, ensure you are inside the target database:

```
use school
```

Command to delete database

```
db.dropDatabase()
```

Response:

```
{ "ok" : 1, "dropped" : "school" }
```

Verify

```
show dbs
```

Now `school` will no longer be listed.

10. Switching Between Databases

Example

```
use admin
use config
```

This is done using the **same use command**.

11. Summary of Key Commands

Database-related commands

Purpose	Command
Show all databases	<code>show dbs</code>
Create/switch database	<code>use databaseName</code>
Show current database	<code>db</code>
Delete database	<code>db.dropDatabase()</code>

Collection-related commands

Purpose	Command
Show collections	<code>show collections</code>
Create collection	<code>db.createCollection("name")</code>
Rename collection	<code>db.oldName.renameCollection("newName")</code>
Delete collection	<code>db.collectionName.drop()</code>

Help commands

Purpose	Command
General help	<code>help</code>
Database help	<code>db.help()</code>
Collection help	<code>db.collectionName.help()</code>
